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Clearing Amyloid-β through PPARγ/ApoE Activation by Genistein is a Treatment of Experimental Alzheimer's Disease

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Abstract. Amyloid-β (Aβ) clearance from brain, which is decreased in Alzheimer's disease, is facilitated by apolipoprotein E (ApoE). ApoE is upregulated by activation of the retinoid X receptor moiety of the RXR/PPARγ dimeric receptor. Genistein, a non-toxic, well-tested, and inexpensive drug activates the other moiety of the receptor PPARγ. Treatment of an Alzheimer's disease mouse model with genistein results in a remarkable and rapid improvement in various parameters of cognition, such as hippocampal learning, recognition memory, implicit memory, and odor discrimination. This is associated with a lowering of Aβ levels in brain, in the number and the area of amyloid plaques (confirmed *in vivo* by positron emission tomography) as well as in microglial reactivity. Finally, incubation of primary astrocytes with genistein results in a PPARγ-mediated increased release of ApoE. Our results strongly suggest that controlled clinical trials should be performed to test the effect of genistein as treatment of human Alzheimer's disease.

Keywords: Aging, biomedical research, neurodegenerative disease, neuroscience, phytoestrogens

INTRODUCTION

Amyloid-β (Aβ) deposition is one of the most critical signs of Alzheimer's disease (AD). The most frequent form of AD is the sporadic one that occurs late in life and is associated with the deposition of Aβ in the brain. It is important to realize that the increased brain levels of Aβ are not due to an elevated rate of synthesis, but rather to a lower clearance from brain [1]. Lowering Aβ levels results in fewer plaques that are responsible for the initiation of synaptic function impairment and finally neuronal death. Thus, ApoE-mediated clearing of Aβ from brain is a major therapeutic target to treat AD [1–3]. Brain ApoE is released mainly from astrocytes and its transcription is activated by the dimerization of the retinoid X receptor (RXR) with peroxisome proliferator-activated receptor gamma (PPARγ) [4]. Thus RXR-PPARγ agonists such as glitazones [5] were promising candidates to activate ApoE synthesis and potentially to treat AD, but attempts to treat AD with these compounds have failed to treat human AD [6] probably because of their poor blood-brain barrier penetration [7]. More recently, bexarotene which binds to the RXR subunit of the

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receptor has been shown to lower Aβ plaque area and rapidly reverse cognitive, social, and olfactory deficits in AD mouse models [8]. However, it has serious undesirable side effects such as promoting hypothyroidism [9, 10] increasing plasma triglycerides [11], and affecting liver function [12]. Genistein, which binds to the PPARγ moiety of the RXR/PPARγ dimer receptor [13–16], is extensively used in clinical practice, for instance to prevent menopause-associated hot flushes [17] and is devoid of significant side effects [18, 19]. We hypothesized that genistein could be a useful drug to treat experimental AD. Thus, we determined the effect of treatment with genistein, bexarotene, or the combination of both, on behavioral, histological, and molecular parameters in the amyloid-β precursor protein/presenilin 1 double transgenic (AβPP/PS1) mouse model of AD.

We found that treatment of AβPP/PS1 AD mice with genistein results in a remarkable and rapid improvement in various parameters of cognition. This is associated with a lowering of Aβ levels in brain, in the number and the area of amyloid plaques as well as in microglial reactivity. Here we show that the beneficial effects of genistein are mediated by a PPARγ–dependent increase in ApoE release from astrocytes. Our results strongly suggest that controlled clinical trials to test the effect of genistein as treatment of human AD should be performed.

METHODS

Animals and treatments

We used the AβPPswe/PS1dE9 [20] mice as model of AD (hereafter referred to as AβPP/PS1) that originates from the C57BL/6J strain. Animals were obtained from Jackson laboratories (USA) (http://jaxmice.jax.org/strain/004462.html) and raised in the animal facility of the Faculty of Medicine, Universitat de València. Animals were housed with *ad libitum* access to water and food, at constant temperature and humidity, with a 12 h light/12 h dark cycle.

All animals used in this work in all cases were 8-month-old females that were ovariectomized, as described in Lopez-Grueso [21], to prevent the effect of endogenous estrogens. Mice were distributed in the following experimental groups: Wild type mice with the same genetic background as the AβPP/PS1 mice; AβPP/PS1 mice and AβPP/PS1 mice treated with bexarotene during 3 days (100 mg/kg/day, as previously described), with a combination of genistein and bexarotene, 3 days of genistein (0.022 mg/kg/day) followed by 3 days of genistein plus bexarotene or with genistein alone during 6 days.

The methods were carried out in "accordance" with the approved guidelines and the different experimental designs were approved by the ethics committee of the University of Valencia (codes: A1348072223268, A1361200660910, and A1389269795198), following the animal care guidelines of the European Commission 2010/63/UE.

Behavioral analysis

Object recognition test

The novel object recognition was used to assess recognition memory [22] and was performed as described previously [23, 24]. The apparatus consisted of an open box (24 × 24 × 15 cm) located in a testing room with constant illumination. The objects used were two small river stones (A) and a small non-toxic plastic toy (B), heavy enough to prevent displacement. On the day before the test, the habituation day, mice were allowed to explore the box (with no objects) for 2 min. On the day of the test, a training session (T1) was followed by a test session (T2) after a 1-min interval. Each session (T1 and T2) lasted 3 min.

For T1, mice were placed in the middle of the box faced away from the two identical stones (AA) arranged in the center of the testing box for 3 min. After this, mice were removed from the box and returned to their home cages. One of the stones was changed to one small toy (non-familiar object). After the retention interval of 1 min outside the testing box, mice were reintroduced into the box for T2. Object exploration was defined as the orientation of the animal's snout toward the object, within a range of 2 cm or less from the object. Running around the object or sitting on it was not recorded as exploration. Objects were washed with ethanol after each individual trial to eliminate olfactory cues. The basic measures in the object recognition test were the times spent by the mice to explore an object during T1 and T2 [24]. *e1* and *e2* are measures of the total exploration time of both objects during T1 and T2, respectively. *d1* was considered as an index measure of discrimination between both the new and familiar objects. Thus, *d1* is the difference in exploration of the two objects in T2.

Passive avoidance test

To determine implicit memory, we used the passive avoidance test. To this end, a step-through inhibitory avoidance apparatus for mice (UgoBasile, Comerio-Varese, Italy) was employed. This cage is made of Perspex sheets and divided into two compartments (15 × 9.5 × 16.5 cm each one). The safe compartment is white and illuminated by a light fixture (10 W) fastened to the cage lid, whereas the "shock" compartment is dark and made of black Perspex panels. The two compartments are divided by an automatically operated sliding door at floor level. The floor is made of 48 stainless steel bars with a diameter of 0.7 mm and situated 8 mm apart.

Passive avoidance tests were carried out essentially following the procedure described in Aguilar et al. [25]. On the day of training, each mouse was placed in the illuminated compartment facing away from the dark compartment. After a 60 s period of habituation, the door leading to the dark compartment was opened. When the animal had placed all four paws in the dark compartment a foot-shock (0.5 mA, 3 s) was delivered and the animal was immediately removed from the apparatus and returned to its home cage. The time taken to enter the dark compartment (step-through latency) was recorded. Retention was tested 24 h and one week later following the same procedure but without the shock. The maximum step-through latency was 300 s.

Hebb-Williams maze

The hippocampal learning was evaluated using the Hebb-Williams maze. The one we used in our experiment is made of black plastic (60 × 60 × 10 cm). It contains a start box and a goal box (both 14 × 9 cm), which are positioned at diagonally opposite corners. The maze contains cold water at a wading depth (15°C, 3.5 cm high), while the goal box is stocked with fresh dry tissue. Several maze designs are produced by fixing different arrangements of barriers to a clear plastic ceiling. This apparatus allows the cognitive process of routed learning and the motivation of water escape to be measured.

The procedure we followed was based on that employed by Galsworthy et al. [26], in which mice must navigate the maze and cross from the wet start box to the dry goal box in order to escape the cold water. The practice mazes were as follows: animals underwent a 5 min habituation period (dry sand, no barriers) on day 1 and undertook problem A on day 2 and problem D on day 3 (4 trials/day) (the description of the problems and all maze designs is in Rabinovitch and Rosvold [27]. Mice were subsequently submitted to mazes 1 and 5 (considered as easy and difficult mazes) on separate days on which 8 trials took place. The time limit for reaching the goal box was 300 s, after which the mouse was guided to the box. The total latency score (the sum of the latencies in all the problem trials in each maze) was recorded. Individuals unable to complete the task within the time limit scored maximum latencies (300 s).

Odor habituation test

Olfactory deficits were screened following the procedure of Wesson et al. [28] using the odor-habituation test [29]. Odors ($n = 3$; heptanone, ethyl valerate and pentanone; Sigma Aldrich, St. Louis, MO) were diluted 1×10^{-3} in mineral oil and applied to a cotton-applicator stick. To prevent contact of the liquid odor with the animal, the stick was enclosed in an odorless plastic tube. This control method of odor presentation permits the minimization of visual and somato-sensory influences. Animals were tested in their own home cage to minimize potential influences of stress or anxiety on the behavioral measures. Testing took place during the dark phase of the animals' day. Odors were delivered for 4 successive trials (1 block), 20 s each, separated by 30-s inter-trial intervals, by inserting the odor stick into a port on the top of the animal's home cage. The number of approaches, defined as snout-oriented sniffing within 1 cm of the odor presentation port was registered. Odor presentation and mice were tested in a counter-balanced order.

For analysis of olfactory behavior data, odor investigation durations within individual trials were collapsed across all odors. As a measure of novel odor investigation, the durations of all trial 1–4 odor investigations were analyzed. As a measure of odor habituation, the investigatory values in the four trials were analyzed in the different treatment groups. To calculate habituation index, the normalized investigatory values from all 4th trial odor presentations were subtracted from the corresponding 1st trial odor presentations.

Histological analysis

AβPP/PS1 mice brains were freshly dissected and one hemisphere from each brain was used for histological analysis while the other one was

intended for molecular assays. Tissue was fixated by overnight immersion in 2% paraformaldehyde, 2.5% glutaraldehyde (EMS, Hatfield, PA, USA), in 0.1 M phosphate buffer (pH 7.4; PB). Coronal 200 μm sections were cut with a vibratome (VT-1000, Leica, Wetzlar, Germany), and were subsequently post-fixated by immersion in 2% osmium tetroxide, rinsed with water, sequentially dehydrated with increasing concentrations of ethanol, and finally embedded in Durcupan™ ACM (Fluka, Buchs, Switzerland). Semithin sections (1.5 μm) were cut with a UC-6 ultramicrotome (Leica), mounted on gelatin-coated slides and stained with 1% toluidine blue. These sections were examined under a light microscope (Eclipse E800; Nikon, Japan). Amyloid plaques are easily identified in semi-thin sections as accumulations of heterogeneous material surrounded and infiltrated by activated microglia. Quantification of amyloid plaques in the cortex was performed from the corpus callosum to the brain surfacein 2-3 different levels per animal, and the average among sections was expressed as number of plaques/mm^2 for each animal. Moreover, amyloid plaques were manually outlined and measured to determine amyloid load (defined as the sum of amyloid plaque area/total area x 100), which was calculated for each section using Fiji [30] and then averaged among sections for at least 5 mm^2 of total area per animal. Data were collected and verified by two independent investigators, blinded to groups, and only structures bigger than 15 μm in diameter where identified as amyloid plaques.

Immunohistochemistry for ionized calcium-binding adapter molecule 1 (Iba1), a microglia/macrophage-specific marker, was carried out in similar brain levels. Briefly, sections were washed in 0.1 M PB and incubated in blocking solution for 1 h at room temperature, followed by overnight incubation at 4ºC with primary antibody rabbit anti-Iba1 (1:300; Wako, Osaka, Japan). Then, sections were washed and incubated with biotinylated anti-rabbit IgG (1:200; Vector Labs, Burlingame, CA, USA) followed by an incubation with avidin-biotin-peroxidase complex (Elite ABC complex, 1:100; Vector Labs), treated with 3, 3'-diaminobenzidine (DAB, 0.05% in Tris-HCl, pH 7.6; Sigma-Aldrich, Switzerland) in the presence of 0.005% H_2O_2, and later visualized under an Eclipse E800 light microscope (Nikon, Japan). Fiji imaging software was used to set a constant intensity threshold for automated DAB selection and quantification of the percentage of area stained.

Determination of amyloid-β concentration

Aβ concentration in total brain was measured by an Aβ ELISA kit (Invitrogen, Camarillo CA, USA). Sample preparation, processing, and detection were performed according to the manufacturer's instructions.

Positron emission tomography (PET) analysis

PET scans were performed on 7 animals (3 non-treated and 4 treated with genistein) after being injected with [^{18}F]-Florbetapir. Animals were anesthetized with an induction chamber using isoflurane (1.5%–2% in 100% oxygen, IsoFlo; Abbott Laboratories). Anesthesia was maintained with an orofacial mask and the same anesthetic at a concentration of 1.5% in O_2. The radiotracer dose was injected in the lateral tail vein. After administration, animals were allowed to recover and were returned to their cages. After the uptake time, which was 40 min for [^{18}F]-Florbetapir, animals were anesthetized again and placed into the animal dedicated PET camera (Albira small animal PET, ONCOVISION, GEM-Imaging) for 10 min. During data acquisition, anesthesia was maintained with an orofacial mask and isoflurane at 1.5% vaporized in O_2. Regions of interest were manually drawn over the brain with PMOD software. Tracer uptake by the brain was quantified as the standard uptake value ratios.

Cell culture

Cortical astrocytes were cultured from dissected cerebral cortices of neonatal mice (C57BL/6J). The cell suspension was seeded in 150 cm^2 culture flasks (NUNC, Roskilde, Denmark) at a density obtained using 3-4 brain cortex per flask. The culture medium was changed twice a week until they reached confluency. Afterwards, they were seeded in 25 cm^2 culture flasks at a density of 500000 cells per flask, and treated after 5-6 days culture (80–90% confluency) with vehicle (DMSO), genistein 1, 2.5, 5, or 10 μM for 24 h, and then Aβ (5 μM) was added to the media for further 24 h. Where indicated in Fig. 4, T0070907, a specific inhibitor of PPARγ [31] was added at 10 nM for 24 h. We collected the supernatants for ApoE determination.

Western blotting

Supernatants form treated astrocytes were used to determine ApoE by western blotting. The pro-

teins present in the samples were separated on a 10% acrylamide gel, and were electroblotted to a PVDF membrane. We proceeded to the incubation of the membrane with the antibody to Antilipoprotein E antibody: Abcam (ab83115). Then developed using the computer program "Imagequant LAS 4000" (GE-HealthcareBio-Sciences), and analyzed with the software "Image J".

Statistical analysis

For behavioral analysis, data relating to the elevated plus maze, the object recognition test and open field were analyzed by a one-way ANOVA, Treatment with five levels. The passive avoidance test was analyzed by a two-way ANOVA, with the same between variable, and one within variable with three levels: training day, 24 h test day and 7 days test.

The data of the Hebb-Williams maze were analyzed by a two-way ANOVA with one between subject variable Treatment, and one within subject variable maze, with 2 levels: Maze 1 (easy maze) and Maze 5 (difficult maze).

The novel odor investigation and habituation index were analyzed using a mixed ANOVA with one between subject variables Treatment. Habituation to odor discrimination was analyzed using a mixed ANOVA with the between variable Treatment and a within variable trial, with four levels: 1st, 2nd, 3rd, and 4th trial. Bonferroni adjustment was employed for *post hoc* comparisons.

For histological and molecular analyses, a one-way ANOVA with one variable (treatment) was used.

Statistical significance was set at $p \leq 0.05$. All results are expressed as mean ± S.E.M.

RESULTS

Genistein improves memory and cognition in AβPP/PS1 AD mice

Deficits in memory and learning in AβPP/PS1 mice have been previously described [5]. We observed that recognition memory, determined by the novel object recognition test [22] was impaired in AD when compared with wild type mice. Treatment with genistein, bexarotene, or both restored it to normal values (Fig. 1a). Hippocampal learning was evaluated using the Hebb-Williams Mazes 1 (easy test) and 5 (difficult test). Results from both tests showed that AβPP/PS1 mice treated with genistein, bexarotene, or the combination of both, needed a shorter time to reach the goal than the untreated AβPP/PS1 mice (Fig. 1b). We also evaluated implicit memory using the passive avoidance test. Seven days after the first training session AβPP/PS1 mice employed a significantly lower time to enter into the dark compartment than wild type mice. This indicates that memory in AβPP/PS1 mice is, as expected, impaired. Loss of memory is prevented by genistein, bexarotene, or both (Fig. 1c).

Loss of the sense of smell is a very early sign of AD [32]. We thus determined the odor discrimination index and found that AβPP/PS1 mice display a significantly lower habituation index to odors than wild type animals, whereas treatment with genistein, bexarotene, or both prevented it (Fig. 1d).

Finally, we have observed that improvements in cognitive performance that we have just described are not due to changes in anxiety measured by the elevated plus maze) or in motor activity measured by the open-field test.

Genistein lowers brain Aβ deposition in AβPP/PS1 AD mice

The presence of amyloid plaques is a hallmark of AD. They are indeed present in the brains of our AD mice. Figure 2 shows that genistein, bexarotene, or their combination lowered the number and total area of the plaques (Fig. 2a-d). Interestingly, genistein was at least as effective as bexarotene in decreasing Aβ deposition in brain (Fig. 2i, j).

Another feature of AD is increased microglial activation, probably as a consequence of amyloid deposition [33, 34]. Figure 2 (Fig. 2e, h) shows that treatments significantly microglial density (measured as Iba1). Interestingly, the intensity of Iba1 protein expression by microglial cells was also reduced, which can be interpreted as a reduction of brain inflammation (Fig. 2k). Moreover, in treated mice we found a substantial decrease in the amount of labeled microglial cell expansions and they were less branched than in untreated mice.

To further investigate the lowering of plaque area and numbers at the molecular level, we measured $A\beta_{40}$ levels in whole brain extracts from AD mice and found that genistein, bexarotene, or the combination of both drugs significantly lowers them (Fig. 3a). The same occurred with $A\beta_{42}$ (a more insoluble isoform and thus more abundant in plaques), except that bexarotene alone was not effective (Fig. 3b). We observed that there is a notable correlation between the molecular measurements of $A\beta_{40}$ or $A\beta_{42}$ and

the number and area of amyloid plaques measured in semithin sections (Fig. 3c-f). All mice treated with genistein alone, or in combination with bexarotene, had lower Aβ levels and lower plaque deposition than untreated ones. Thus we show that genistein treatment lowers both Aβ levels and plaque deposition in the brain of AD mice.

Using [^{18}F]-Florbetapir, a specific marker of Aβ [35], we assessed amyloid burden by PET. Whole brain mean florbetapir standard uptake value ratios were lower in genistein treated AβPP/PS1 mice than in untreated ones (Fig. 3g, h). These results confirm the histological and molecular findings *in vivo*. Using PET technology with [^{18}F]-Florbetapir is very interesting because it offers a safe tool to evaluate the Aβ deposition and its progression *in vivo*, without toxicological effects [36]. The fact that the genistein effect can be detected by PET is important for the translation to humans.

As expected, Aβ levels in blood did not change significantly with the treatments (untreated AβPP/PS mice 210 ± 42; genistein treated 320 ± 90 pg/mL of blood). This can be explained by the well- known active hepatic degradation of plasma Aβ.

The mechanism underlying the genistein effect can be explained, at least in part, because it increases ApoE release from astrocytes. Treating mouse primary astrocytes with genistein increases ApoE release to the culture medium in a concentration dependent manner (Fig. 4a, b). This effect is mediated by activation of PPARγ as its inhibition significantly prevents the increase in genistein- induced ApoE release (Fig. 4c).

DISCUSSION

We report here the beneficial effect of genistein, bexarotene, or both on brain Aβ load, cortical plaque number and size, brain microglial inflammation, and important cognition parameters, such as hippocampal learning, implicit memory, and odor discrimination in AD mice. Moreover, PET confirmed that genistein significantly lowers the amount of Aβ plaques in brains of the mice *in vivo*. This can be explained by

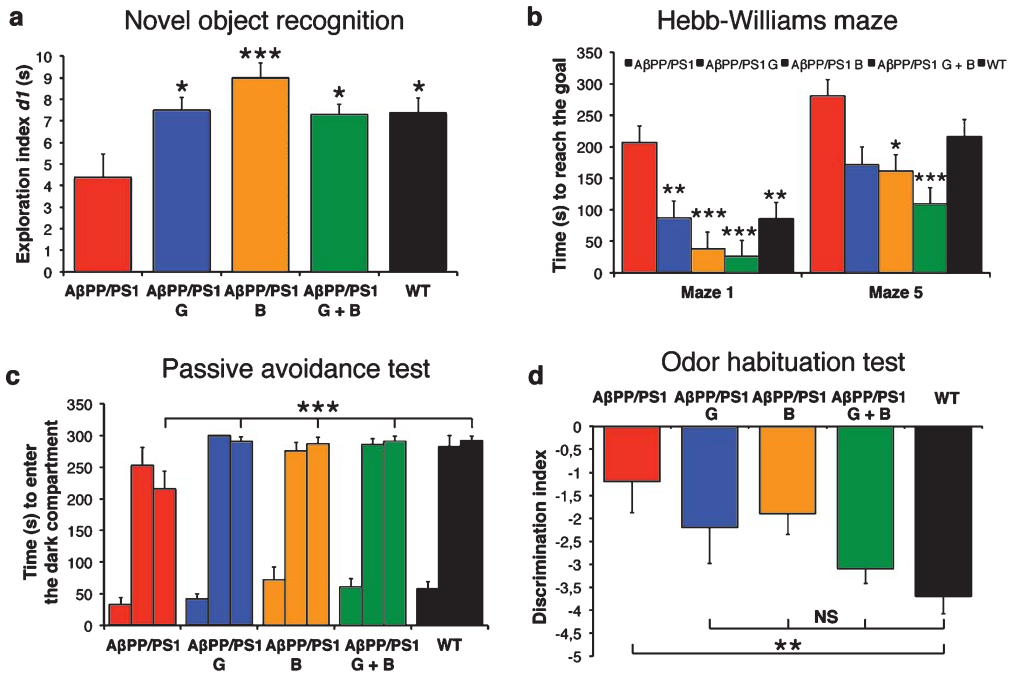

Fig. 1. Genistein restores cognition and memory in AβPP/PS1 mice. a) Recognition memory was studied using the novel object recognition test: The discrimination index *d1* was 4.4 ± 0.8 s in AβPP/PS1 mice, and returned to wild-type (WT) mouse values (7.4 ± 0.6 s), in genistein (G), bexarotene (B), or genistein plus bexarotene-treated mice (G+B). b) Hippocampal learning was evaluated using the Hebb-Williams maze. Genistein, bexarotene, or both restored hippocampal learning impairment in the easy test (maze 1) and in the difficult test (maze 5). In all cases, AβPP/PS1 mice took longer time to reach the goal than WT or treated mice. c) Implicit memory was assessed using the passive avoidance test. All groups presented longer step-through latencies in both 24-h and 1-week test sessions than in the training session. However, AβPP/PS1 mice took less time to enter the dark compartment after 7 days than WT or treated mice. d) Olfactory deficits were assessed by the odor habituation test. AβPP/PS1 mice's discrimination index was lower than that of WT. Treated mice did not show a significantly lower discrimination index than that of WT mice. Data are expressed as Mean ± SEM. Significance is shown as $^{*}p \leq 0.05$; $^{**}p \leq 0.01$; $^{***}p \leq 0.001$, $n = 9$–11 animals per group.

our finding that genistein caused a PPARγ- dependent increase of ApoE secretion from astrocytes.

The beneficial effect of bexarotene on experimental AD was previously described by Cramer and co-workers [8]. Similar results had been obtained by Fitz et al. [37] and by Veeraraghavalu et al. [38]. We also confirm these results; however, other authors were not able to see a bexarotene-induced lowering of Aβ levels [37, 39, 40]. This may be due to strain differences or the bexarotene way of administration.

Regarding genistein, a previous report showed that very high doses could improve cognition [41]. These authors injected Aβ into the brain of rats and observed that administration of genistein (10 mg per Kg body weight, oral) inhibited Aβ aggregation and improved cognition after surgery [41]. An important setback of these studies is that brain injection of Aβ is not a good model of AD because it is acute and because metabolic and neurochemical mechanisms are unlikely to be similar to those found in the human disease. We used a much lower dose of genistein (0.02 mg/kg body weight, day) because higher doses inhibit tyrosin kinases activity [19, 42]. Moreover, the dose we used leads to plasma genistein concentrations similar to those found in populations (like Japan) who take substantial amounts of soya—a genistein rich food—in their normal diet [43]. We show that genistein increases ApoE production by astrocytes in culture in a PPARγ-dependent manner, thereby suggesting a mechanism

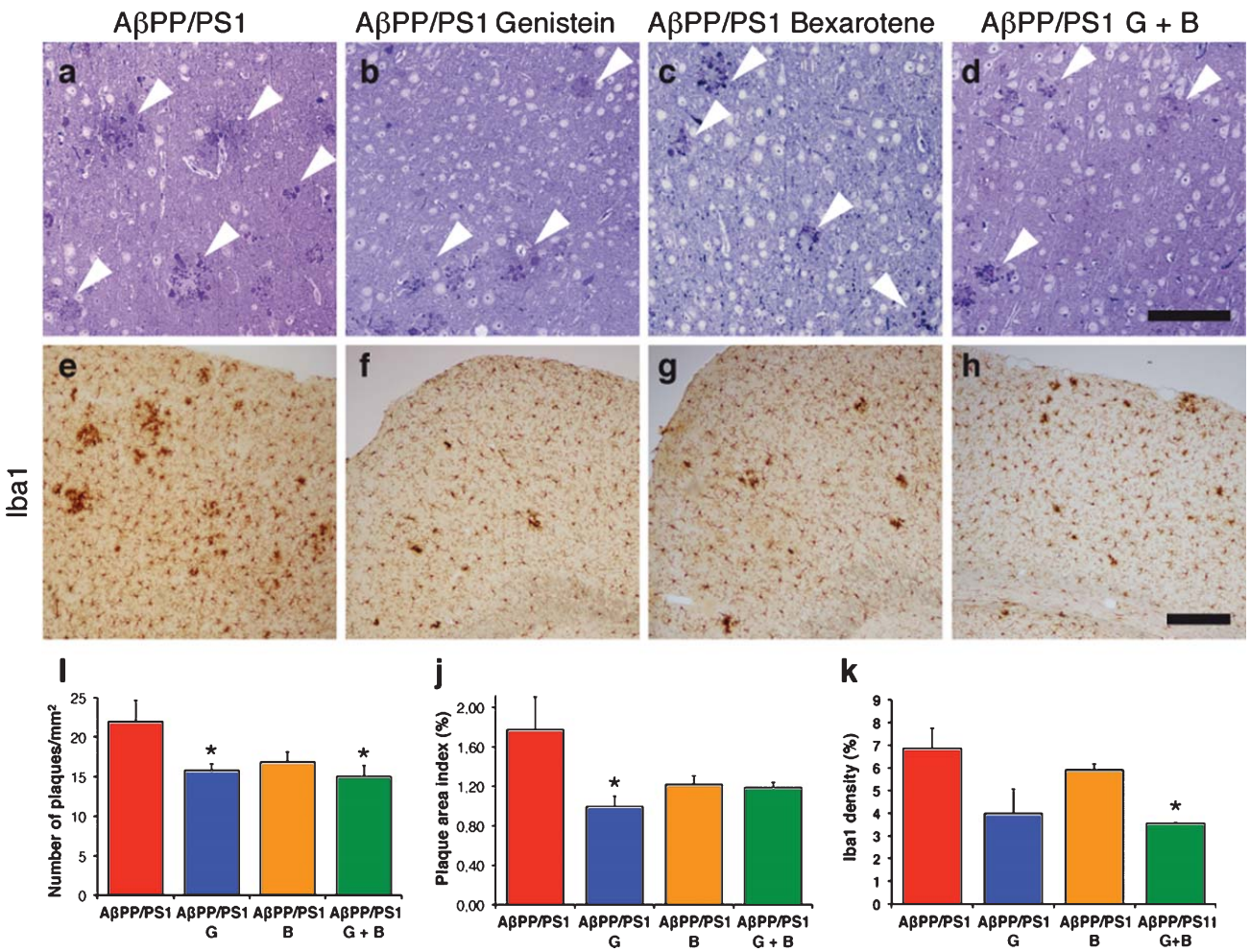

Fig. 2. Amyloid plaques and microgliosis decrease in genistein-treated AβPP/PS1 mice. a-d) Representative image of semithin sections of cerebral cortex show that the number and area of amyloid plaques (arrowheads) are reduced in treated AβPP/PS1 mice when compared to untreated AβPP/PS1 mice. e-h) Microglial activation in cerebral cortex determined by Iba1 immunostaining is reduced in treated AβPP/PS1 mice when compared to untreated AβPP/PS1 mice. Images also show that microglial density associated to amyloid plaques is reduced in treated AβPP/PS1 mice. i-k) Quantitative analysis of the images shows that genistein-treated AβPP/PS1 mice are significantly different from untreated ones. Data are expressed as Mean ± SEM. Significance is shown as $^*p \leq 0.05$, $n = 5$–7 animals per group. Scale bar A-D = 100 μm; E-H = 200 μm.

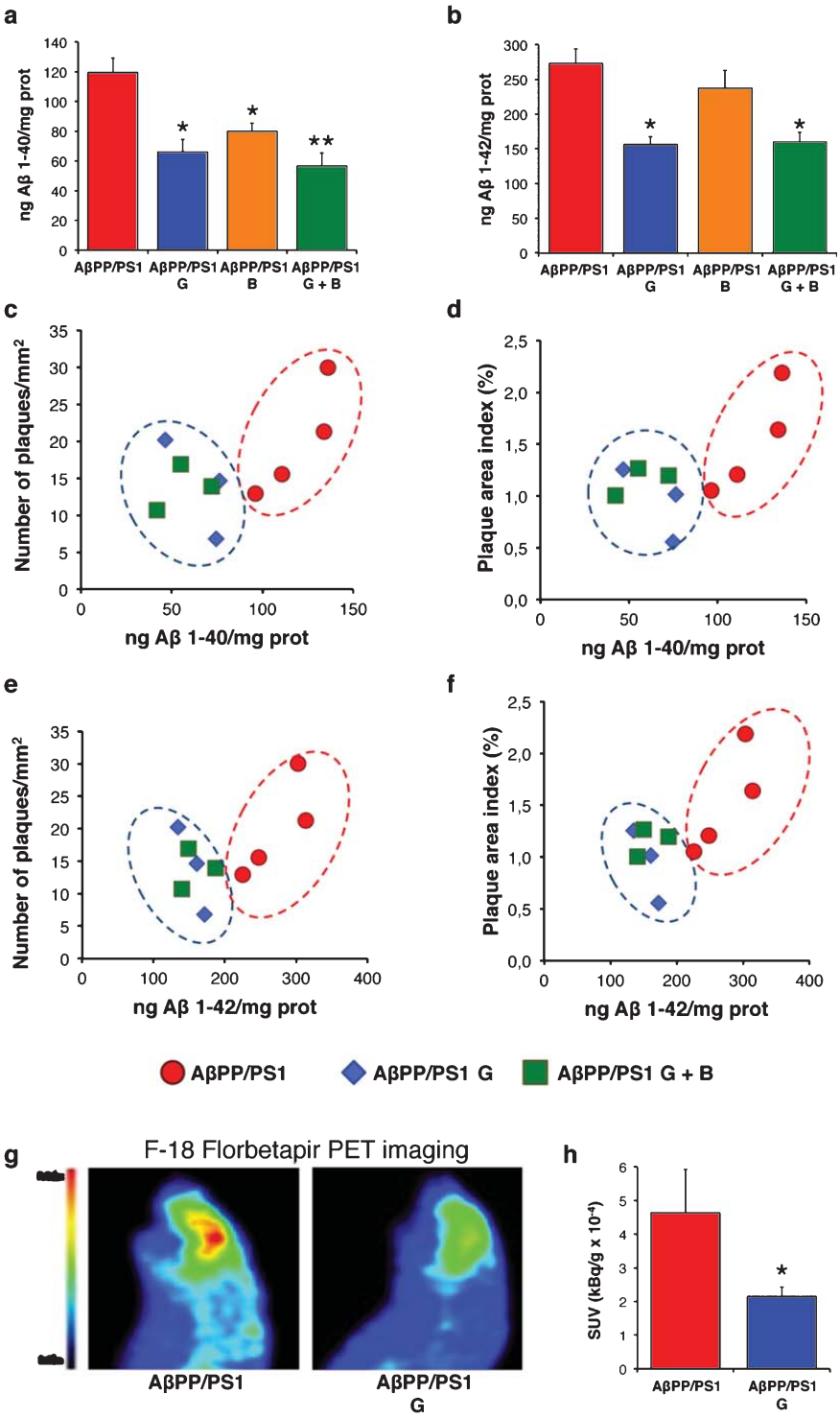

Fig. 3. Genistein lowers amyloid-β (Aβ) levels and plaques in brain. a,b) $A\beta_{40}$ or $A\beta_{42}$ levels (determined in whole brain homogenates by ELISA) are lower in genistein-treated AβPP/PS1 mice than in untreated ones. c-f) Correlation between $A\beta_{40}$ or $A\beta_{42}$ levels in whole brain versus number of plaques/mm^2 or plaques area/total area quantified in semithin sections. Mice are clearly separated in two groups, untreated (highlighted in the red circle) versus genistein-treated (blue circle). g) *In vivo* brain ^{18}F-Florbetapir uptake. Representative image of the sagital view of the brain (the intensity is expressed as maximal uptake per slice) in untreated and genistein treated AβPP/PS1 mice. h) Brain ^{18}F-Florbetapir uptake, expressed as standardized uptake value (SUV). Data are expressed as Mean ± SEM. Significance is shown as $^{*}p < 0.05$, $^{**}p < 0.01$ versus controls; $n = 3$-4 mice.

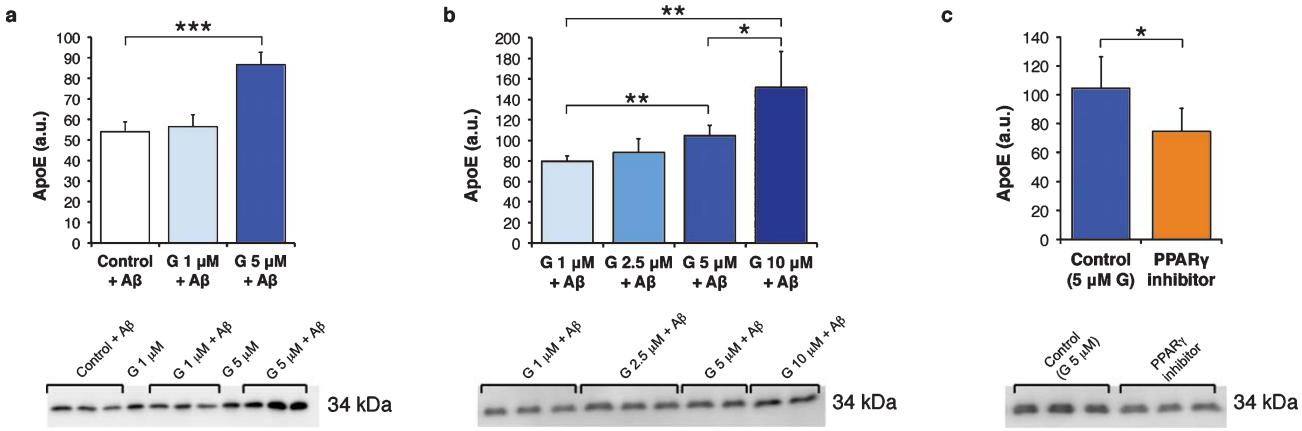

Fig. 4. Genistein increases ApoE release by astrocytes in primary culture mediated by PPARγ. a) ApoE release by astrocytes is increased by 5, but not 1 μM genistein. Data are expressed as Mean ± SEM ($n = 5$-6 experiments). A representative western blot image is shown. b) Concentration course (1–10 μM) of the effect of genistein on ApoE release from astrocytes. The lowest effective concentration tested was 5 μM. Data are expressed as Mean ± SEM ($n = 5$-6 experiments). A representative western blot image is shown. c) Inhibition of PPARγ by T0070907 (10 nM) significantly reduces genistein ApoE-induced release to the medium. Data are expressed as Mean ± SEM ($n = 5$-6 experiments). A representative western blot image is shown. Significance is shown as $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ versus controls.

of action of its beneficial effects in experimental AD (see Fig. 4).

APOE is polymorphic, with three major alleles: ApoE2, ApoE3, and ApoE4. It is well known that APOE isoforms influence cognition and the development of AD, the E4 variant being the largest known genetic risk factor for late-onset sporadic AD in a variety of ethnic groups [44]. It is important to take this E4 isoform-associated risk in consideration when proposing ApoE directed therapeutics, as ApoE4 is not able to clear Aβ. In mice that are not genetically manipulated for these isoforms, there are no ApoE variants, but this factor will be very important for the translation of our results to humans. A major advantage of genistein versus bexarotene is that the latter is an antineoplastic drug with severe side effects, which call into serious question its chronic use in AD patients. Indeed, Tesseur et al. reported that bexarotene-treated mice showed significant weight loss, increased irritability, and difficult breathing [40]. Laferla reported similar concerns based on the severe side effects of the drug [3]. Genistein is devoid of significant toxicity. Its possible use in AD may be reinforced by the epidemiological fact that countries, such as Japan, in which genistein intake is higher than in the West, have considerably less incidence of AD (http://www.worldlifeexpectancy.com/cause-of-death/alzheimers-dementia/by-country/), even after correcting for genetic changes [45].

The strength of this study lies in the fact that we have used very different parameters to evaluate the effect of genistein as treatment of experimental AD such as molecular determination of Aβ levels, histological and PET analysis, and a complete set of behavioral tests. In all cases, genistein showed a remarkably positive effect. Finally, we also propose a molecular mechanism for this effect.

The clinical relevance of this study is that the clear favorable effects of genistein, together with its lack of undesirable side effects, even accepting the major limitation of using a murine model of AD, strongly suggest that controlled clinical trials should be performed to test the effect of genistein as treatment of human AD.

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